Harsh Mali

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Education

The Pennsylvania State University — University Park, PA
Honors Bachelor of Science in Mechanical Engineering; GPA: 3.89/4.00 — *Aug. 2025 – May 2028*

Relevant Coursework: Dynamics, Engineering Thermodynamics, Strength of Materials, Statics

Experience

Research Assistant — Jan. 2026 – Present
S.H.A.P.E. Lab — *University Park, PA*

- Developed Python image-processing workflows to analyze high-speed synchrotron X-ray imaging of multi-material laser powder bed fusion, supporting studies of spatter and melt-pool instability.
- Co-authored research on Cu10Sn bronze and 316L stainless steel using in-situ X-ray imaging and CFD simulations to investigate spatter formation and interfacial material mixing.
- Initiated thesis research combining Python and CAD model analysis to characterize and predict the geometry of 3D-printed sand-casting molds.

Project Lead & Former Manufacturing Lead — Aug. 2025 – Present
American Society of Mechanical Engineers — *University Park, PA*

- Led manufacturing and assembly of a remotely operated waste-collection robot, translating design requirements into a build that met competition size, safety, and performance constraints.
- Contributed to a 3rd-place national finish at the EFx Competition through robot design, fabrication, and testing.
- Lead the current EFx robotics team, coordinating project planning, technical development, and manufacturing across subteams to integrate their work into the overall robot design.

Payload Member — Jan. 2026 – Present
Nittany Rocket Labs — *University Park, PA*

- Designed and integrated an electronic payload system to collect, process, and store in-flight data for the International Rocket Engineering Competition (IREC).
- Developed and tested embedded software, sensor interfaces, and telemetry systems for real-time data acquisition and transmission to a ground station.
- Support flight preparation through system integration and validation testing of payload electronics and avionics.

Mentor — Apr. 2026 – Present
Undergraduate Research and Fellowships Mentoring — *University Park, PA*

- Guide undergraduates seeking research and fellowships through faculty outreach, application strategies, and campus resources, helping them navigate available university opportunities.

Projects

Reaction Wheel Balance Controller | *Arduino, C++, IMU* — Jan. 2026 – May 2026

- Developed a single-axis self-stabilizing platform using a reaction wheel and closed-loop feedback control to maintain autonomous balance.
- Integrated mechanical components with an Arduino, motor driver, and MPU6050 IMU to connect sensor feedback with motor actuation.
- Validated platform performance through iterative testing and controller tuning.

Technical Skills

Languages: Python, MATLAB, C++, HTML
CAD & Simulation: SolidWorks, Autodesk CFD, Fusion 360, Onshape, Inventor, Inventor FEA, N-Top
Embedded Systems: Arduino, embedded C++, circuit prototyping, sensor interfaces, IMUs, motor drivers, telemetry
Productivity: Microsoft 365 (Excel, Word, PowerPoint)